Yale JACKSON SCHOOL OF GLOBAL AFFAIRS

Assignment 6: Analyzing Gains from Trade **Due Sunday, March 29**

GLBL 5010 Economics for Global Affairs

This assignment asks you to perform a simple analysis to measure the gains from trade using data on domestic expenditure shares and estimates of the trade elasticity. We will be using the World Input-Output Database (WIOD) tables from 2000 to 2014. This analysis provides some insights into the following questions: How much has a country's economy benefited from trade? Can most of the trade gains be traced to certain countries? How much do the US, Canada, Mexico, and China stand to lose if they become more closed?

We will first start by discussing the theory behind the exercise.

Part I: Another Way to View Openness: λ_{ii}

In class, you are exposed to λ_{in}, the expenditure share of goods that i purchases from n. We have the following identity:

$$\lambda_{in} = \frac{(p_{in})^{-\epsilon}}{\sum_k (p_{ik})^{-\epsilon}} = \frac{\left(\frac{w_n}{A_n} d_{ik}\right)^{-\epsilon}}{\sum_k \left(\frac{w_k}{A_k} d_{ik}\right)^{-\epsilon}}$$

1. Based on the identity above, explain in your own words what happens to the expenditure share λ_{in} if (i) productivity of a country increases and (ii) distance changes.

2. ϵ is the elasticity of trade flows. What happens to λ_{in} if trade costs increase? Explain how the answer relates to ϵ and its expected sign.

3. Explain in your own words why the equation $((\lambda_{ii})^{-1/\epsilon} - 1) * 100$ can be used to measure the gains from trade.

Part II: Measuring Gains from Trade

The following questions can be solved using the excel sheet provided that contains the λ_{ii} (Domestic_ shares.xlsx) for each country across time. You can use any data tools (excel, Stata, R, AI) to answer the questions. Hints on how to use the WIOD tables are provided in the end.

4. Fill in the following table below. The table reports the gain from trade relative to autarky.

Country	λ_{ii} (2000)	Gains from Trade (2000)	Gains from Trade (2000)	λ_{ii} (2014)	Gains from Trade (2014)	Gains from Trade (2000)

		$\epsilon = 4$	$\epsilon = 5$		$\epsilon = 4$	$\epsilon = 5$
United States						
Canada						
Mexico						
China						

5. Based on your answer above, how do the gains from trade change when ϵ changes? Explain intuitively why this makes sense.

6. We will now calculate how the gains from trade have changed from 2000 to 2014 using $\epsilon = 5$. To start, fill in the tables below with λ_{ii} and λ_{in} just for the US. Other numbers are provided for you so you can check. You will need to use the WIOD table from 2014.

2014 to/from	**US**	**China**	**Mexico**	**Canada**	**ROW**
US	xxx	xxx	xxx	xxx	xxx
China	0.004	0.941	0.000	0.001	0.055
Mexico	0.084	0.018	0.830	0.004	0.064
Canada	0.090	0.015	0.006	0.832	0.057

2000 to/from	**US**	**China**	**Mexico**	**Canada**	**ROW**
US	0.932	0.003	0.007	0.012	0.046
China	0.004	0.931	0.000	0.001	0.064
Mexico	0.097	0.002	0.845	0.003	0.054
Canada	0.114	0.004	0.005	0.816	0.061

7. Using the tables above, calculate the "Total Gains from Trade" and decompose the contribution and fill in the tables below. Parts of the table are filled for you to check.

Country	Total Gains from Trade (2000 – 2014)	Contribution: trading with the US	Contribution: trading with Canada	Contribution: trading with Mexico	Contribution: trading with China	Contribution: trading with Rest of the World
United States	xxx	NA	xxx	xxx	xxx	xxx
Canada	-0.0039	1.480	NA	-0.057	-0.721	0.299
Mexico	0.0035	-0.885	0.086	NA	1.090	0.709
China	-0.0020	0.030	0.018	-0.009	NA	0.961

8. Write two to three paragraphs based on the results of the two tables above. You can add additional analysis if you want. To receive full points, relate your paragraphs to one or more

questions asked at the beginning of the assignment and you will need to explain what at least one of the numbers mean. Questions from the beginning:

- *How much has a country's economy benefited from trade?*
- *Can most of the trade gains be traced to certain countries?*
- *How much do the US, Canada, Mexico, and China stand to lose if they become more closed?*

Additional Information: Understanding the WIOD Tables

The assignment 6 folder contains the reference paper by Timmer et al. (2016) that talks about WIOD in more detail. The paper provides Figure 1 below that helps you read the table. Using this table, you will need to select the correct data to calculate λ_{ii}.

			Use by country-industries							Final use by countries			Total use
			Country 1			…	Country M			Country 1	…	Country M	
			Industry 1	…	Industry N	…	Industry 1	…	Industry N		…		
Supply from country-industries	Country 1	Industry 1											
		…											
		Industry N											
	……												
	Country M	Industry 1											
		…											
		Industry N											
Value added by labour and capital													
Gross output													

Figure 1. Schematic Outline of a World Input–Output Table (WIOT)

Hint on how to calculate the total expenditure in the US in 2014 in Stata:

```
drop if Country=="TOT" /* This is so that we don't include it when we sum */

* Keep output columns for USA

keep IndustryCode IndustryDescription Country RNr Year vUSA1 - vUSA56 vUSA57 - vUSA61

collapse (sum) vUSA1 - vUSA56 vUSA57 - vUSA61

egen total_exp_USA = rowtotal(vUSA1 - vUSA56 vUSA57 - vUSA61)

di total_exp_USA

*31451596
```

There is also an example code for Stata to calculate λ_{in} for Australia. One could easily translate this in R using LLMs.